Lec-38

Both are sorted

A $\rightsquigarrow$ [ ] m

B $\rightsquigarrow$ [ ] n

single sorted array (merge)

$TC = \theta(m+n)$
$= O(m+n)$
$= \Omega(m+n)$

$\boxed{\begin{array}{l} \rightarrow \text{min no. of comp}^n \Rightarrow \min(m,n) \\ \rightarrow \text{max " " } \Rightarrow m+n-1 \\ \rightarrow \text{moves} \Rightarrow m+n \text{ each (case)} \end{array}}$

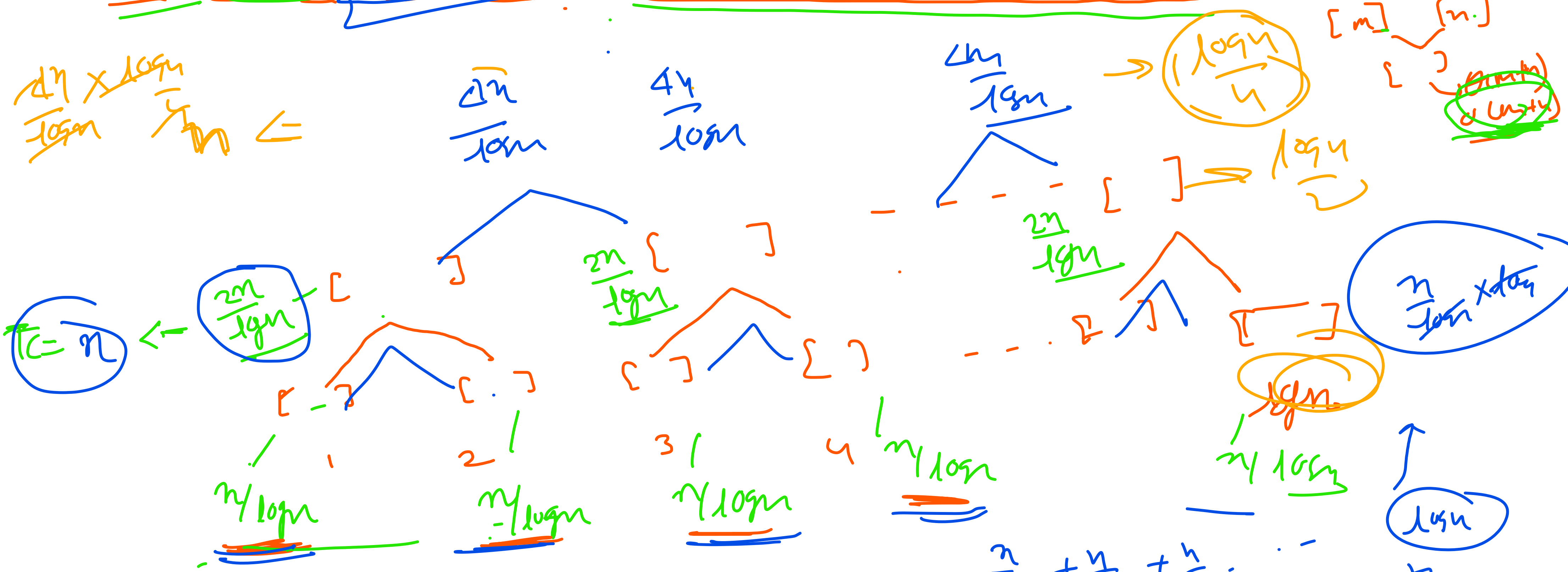
$TC = \theta(m+n)$

Lec_39

Practice Questions on Merge Sort

$n \log \log n$

1.) there are logn sorted subarrays each size n/logn ,find single sorted array with total n elements ,find worst case time complexity.



$\frac{2n}{\log n} + \frac{4n}{\log n}$

$\left(\frac{\log n}{2}\right) 2n \Rightarrow$

$\frac{2n}{\log n} \times \frac{\log n}{2}$

$\frac{n}{\log n} + \frac{n}{\log n} + \frac{n}{\log n} \cdots$

$\frac{\log n}{2^n} = 1$

n

$\Rightarrow 2^n = \log n$

$n = \log \log n$

Time at each level is $\Rightarrow n \quad \Rightarrow n^{\alpha} \quad \Rightarrow n \log \log n$

n

$8n/\log n$

$\log n / 2^3$

$4n/\log n$

$4n/n$

$\frac{4n}{\log n} \times \frac{\log n}{4} \Rightarrow n$

$\frac{4n}{\log n}$

$\log n / 4^2$

$\frac{2n}{\log n} \times \frac{\log n}{2} = n$

$2n/\log n$ $\quad$ $2n/\log$ $\quad$ $4n/n$ $\quad$ $2n/n$ $\quad$ n^{α} $\quad$ $\log n / 2^1$

$\Rightarrow \log n / 2^0$

$n/\log n \quad n/\log n \quad n$

$1 \quad n/16 \quad n/2^4$

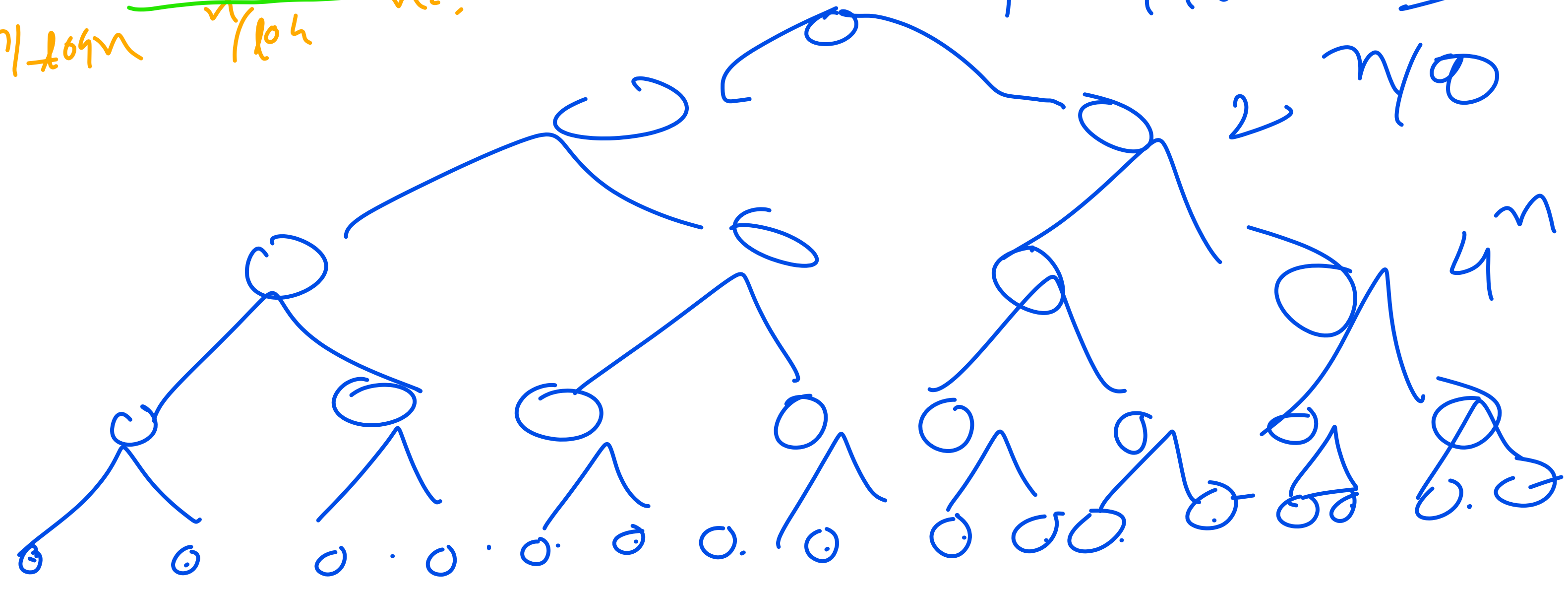
$2 \quad n/8 \quad n/2^3$

$4 \quad n/4 \quad n/2^2$

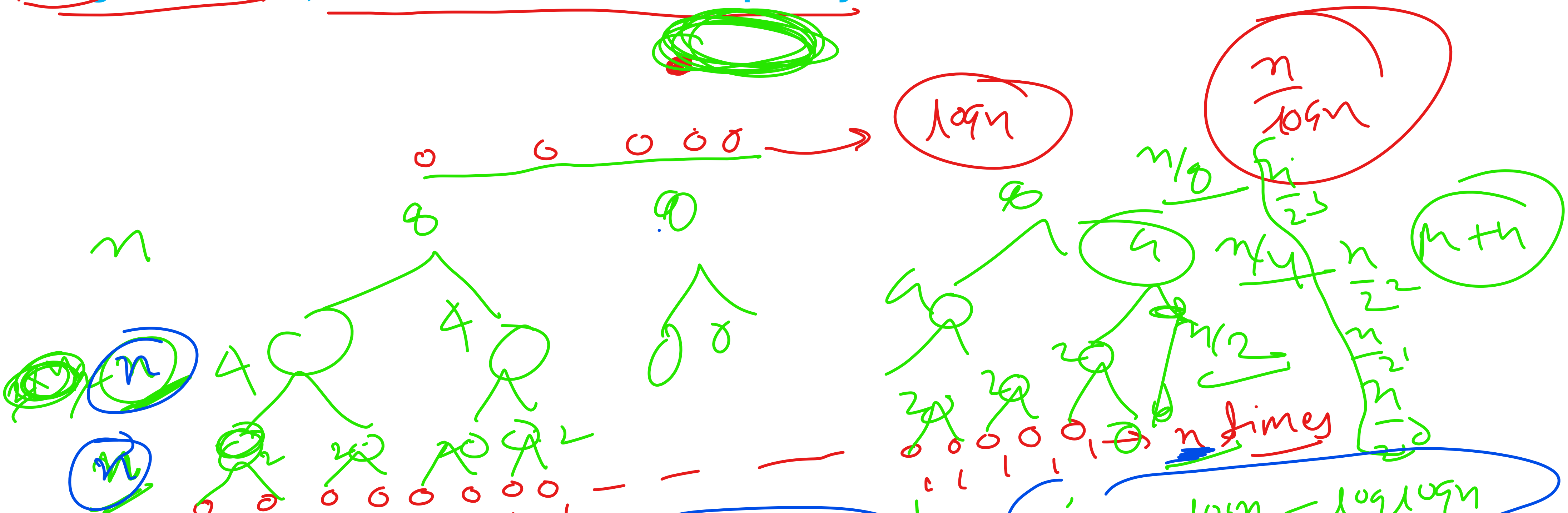
$n/2 \quad n/2^1$

8

$16 \quad n$



4.) there are n sorted subarrays each size 1 ,find logn sorted arrays with each of n/logn elements ,find worst case time complexity.



$\frac{n}{2^x} = \log n$

$\Rightarrow \frac{n}{\log n} = 2^x$

$\Rightarrow \log\left(\frac{n}{\log n}\right) = x$

$\Rightarrow x = \log n - \log\log n$

$TC = nx$

$TC = n(\log n - \log\log n)$

$TC = n\log n$

TC = $n\log n$

practical

small

3 1.) there are logn subarrays each size n/logn ,find single sorted array,find worst case time complexity.